GRI APP

(THE GANDHIGRAM RURAL INSTITUTE OFFICIAL MOBILE APPLICATION PORTAL)

MINI PROJECT

Submitted to

THE GANDHIGRAM RURAL INSTITUTE

(DEEMED TO BE UNIVERSITY)

in partial fulfilment of the award of the Degree of

MASTER OF COMPUTER APPLICATIONS

BY

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THE GANDHIGRAM RURAL INSTITUTE (DEEMED TO BE UNIVERSITY)

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Accredited by NAAC with 'A++' Grade (4th Cycle)

Gandhigram, Dindigul District,

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DECEMBER 2026

BONAFIDE CERTIFICATE

This is to certify that the project titled "**GRI APP – THE GANDHIGRAM RURAL INSTITUTE OFFICIAL MOBILE APPLICATION PORTAL**" is a Bonafide record of work carried out by **VIJAY PRADHAP S (25322020**), submitted in partial fulfillment of the requirements for the award of the degree of Master of Computer Applications (MCA) at The Gandhigram Rural Institute (Deemed to be University), Gandhigram, during the period July 2026 – December 2026.

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Submitted for the Viva-Voce examination held on _____.

INTERNAL EXAMINAR

EXTERNAL EXAMINAR

DECLARATION

I hereby declare that this mini project work titled "**GRI APP - THE GANDHIGRAM RURAL INSTITUTE OFFICIAL MOBILE APPLICATION PORTAL**" is a record of original work done by me under the supervision and guidance of **Dr. P. KALAVATHI, M.C.A., M.Phil., Ph.D., Professor**, Department of Computer Science and Applications. The mini project work has not formed the basis for the award of any Degree/Diploma/Associateship/Fellowship or similar title to any candidate of any other University.

Place: Gandhigram

Date: _____ **(VIJAY PRADHAP S)**

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ABSTRACT

Students, research scholars, and faculty members residing in rural Tamil Nadu and commuting to The Gandhigram Rural Institute (Deemed to be University) often encounter substantial administrative friction when accessing campus circulars, syllabus records, attendance figures, and examination credentials. The university's statutory operations span eight academic Schools and over thirty specialized departments, serving a student community distributed across interior rural hamlets in Dindigul, Madurai, and Karur districts. Erratic cellular connectivity, non-responsive desktop web layouts on smartphones, and the absence of a unified mobile service platform create information gaps, forcing candidates to undertake unnecessary physical travel and endure lengthy administrative queues for basic academic validations.

The Gandhigram Rural Institute Official Mobile Application Portal (GRI APP) is an advanced native Android application designed to provide students, faculty, and administrative authorities with a single, highly intuitive, bilingual (Tamil and English) mobile gateway to university governance. The application establishes an indexed directory mapping all thirty-plus departments, semester curricula, credit frameworks, and over five hundred cataloged statutory circulars and regulations. Built upon modern cross-platform engineering, the interface incorporates the cultural ethos of the university through a dedicated Gandhian aesthetic design system, delivering accessible typography, responsive touch controls, and resilient performance across budget mobile hardware.

The application incorporates a robust Supabase cloud database with Row-Level Security, alongside a dual-tier in-memory fast cache that delivers complete offline operational availability for remote students. Key integrated modules include an institutional user registration and administrative approval dossier, an automated seventy-five percent attendance-gated examination hall ticket generator with ISO/IEC 18004 e-SANAD QR cryptographic verification, statutory UGC-compliant student grievance redressal (GRI-Care), a digital campus transit bus pass generator, and the GRI-Sahayak intelligent on-device conversational assistant for instant academic guidance. The application is engineered using the Flutter framework, Dart 3.5 sound null-safety, and the hardware-accelerated Impeller engine.

CHAPTER 1

INTRODUCTION

1.1 Historical Background and Gandhian Pedagogy

The Gandhigram Rural Institute (GRI) was established in 1956 under the visionary stewardship of two dedicated disciples of Mahatma Gandhi: **Dr. T. S. Soundram** and **Dr. G. Ramachandran**. Situated in the foothills of the Sirumalai range within the Chinnalapatti valley of Dindigul District, Tamil Nadu, the institute was conceived as a living laboratory for Mahatma Gandhi's revolutionary educational philosophy—***Nai Talim*** (Basic Education).

The foundation of *Nai Talim* posits that meaningful education cannot be isolated from productive physical labor, village life, and rural self-reliance. Under this paradigm, academic pursuit extends far beyond theoretical classroom instruction; it actively engages students in grassroots participatory research, village extension services, rural industrialization, and community health. Recognizing this pioneering institutional model, the Government of India, upon the recommendation of the University Grants Commission (UGC), conferred the status of **Deemed-to-be-University** upon the institute in 1976 under Section 3 of the UGC Act, 1956.

Over nearly seven decades of academic excellence, GRI has expanded into a prestigious national centre for higher learning, accredited with an **'A' Grade (3rd Cycle)** by the National Assessment and Accreditation Council (NAAC). Operating as an autonomous university fully funded by the Ministry of Education, Government of India, GRI integrates conventional disciplines with specialized rural disciplines, including Rural Energy Technology, Khadi and Handloom Development, Rural Industries, Public Health, and Advanced Computer Applications.

1.2 Institutional Profile: 8 Schools and 30+ Departments

The academic architecture of The Gandhigram Rural Institute is structured into eight statutory Schools, each presided over by an Academic Dean and encompassing over thirty specialized teaching and research departments:

1. **School of Sciences & Rural Technology:** Department of Computer Science & Applications, Department of Mathematics, Department of Physics, Department of Chemistry, and Department of Rural Technology.
2. **School of Agriculture & Animal Husbandry:** Department of Agriculture, Department of Animal Husbandry, and the ICAR Krishi Vigyan Kendra (KVK).

3. **School of Rural Health & Sanitation:** Department of Public Health, Kasturba Hospital Clinical Training Center, and Sanitation Science Institute.
4. **School of Social Sciences:** Department of Rural Development, Department of Applied Research, and Department of Economics.
5. **School of Management Studies:** Department of Rural Industries & Management, and Department of Co-operative Management.
6. **School of English & Foreign Languages:** Department of English, Language Laboratory, and Center for Translation.
7. **School of Tamil, Indian Languages & Fine Arts:** Department of Tamil, Center for Performing Arts, and Gandhian Cultural Centre.
8. **School of Education:** Department of Education, and Center for Adult Education & Continuing Studies.

In addition to academic faculties, the university maintains centralized administrative institutions: the **Office of the Controller of Examinations (COE)**, the **Dr. Radhakrishnan Central Library**, the **Internal Quality Assurance Cell (IQAC)**, the **Central Computer Center**, and the **Directorate of Distance Education**.

1.3 Problem Statement & Operational Bottlenecks

While over 95% of GRI students, research scholars, and faculty members own and operate personal Android smartphones, university administrative operations historically remained dependent upon legacy desktop websites, physical noticeboards, and paper records. Students commuting daily from interior rural hamlets across Dindigul, Madurai, and Karur districts faced systemic challenges:

- **Administrative Desk Congestion:** During peak academic windows (semester fee payment, attendance verification, and exam registration), hundreds of students congregated outside administrative buildings, resulting in long queues and loss of active lecture hours.
- **Physical Document Vulnerability & Lost Petitions:** Grievance submissions regarding hostel facilities, campus transit, and academic queries were handled through physical paper slips, creating tracking bottlenecks and communication delays.
- **Rural Connectivity Constraints:** Commuters frequently travel through cellular blind spots where 4G/5G connections drop to intermittent 2G or complete disconnection. The

inability of web pages to function offline left students stranded without access to timetables or faculty contact details.

- **Passive Information Dissemination:** Time-sensitive notices (such as severe weather holidays, scholarship deadlines, and exam schedules) were passively posted on the web portal without push notification mechanisms, leading to missed deadlines.

1.4 Technical Audit of the Legacy Web Architecture

An exhaustive engineering audit of the authoritative university domain (ruraluniv.ac.in) uncovered several fundamental technical limitations on modern mobile viewports:

1. **Monolithic Server-Side Scripting:** The existing portal uses legacy query-parameter PHP routing (academics?content=CBCSsystem), forcing complete server re-rendering on every user interaction with zero client-side caching.
2. **Fixed-Viewport Layouts:** Core informational tables (fee schedules, semester timetables, circulars) feature hard-coded widths exceeding 1,000 pixels, producing horizontal overflow, truncated columns, and illegible text on mobile screens.
3. **Unindexed Document Repositories:** Approximately 600 official statutory PDF forms are distributed across scattered directory paths without unified search indexing or metadata tagging.
4. **Absence of Role-Based Personalization:** The portal treats all visitors identically, lacking dedicated student desks, faculty roll-call interfaces, or administrative review portals.

1.5 Objectives of the GRI Mobile Portal Project

To eliminate these administrative frictions, the **GRI Official Mobile Application Portal (gri_app)** was designed and implemented with the following primary engineering objectives:

1. **Establish a Mobile-First Institutional Cockpit:** Provide an identity-aware mobile interface tailored for Students, Faculty, Scholars, and Administrators.
2. **Ensure Complete Offline Availability:** Implement dual-tier local caching allowing critical academic records, departmental contacts, and syllabus catalogs to remain accessible during complete network loss.
3. **Automate Examination Gating & Verification:** Digitally enforce the statutory 75% lecture attendance threshold for examination eligibility and synthesize tamper-evident e-SANAD QR hall tickets.

4. **Streamline Campus Redressal & Services:** Digitize student grievance logging (GRI-Care) under UGC guidelines and issue digital campus bus passes with real-time transit telemetry.
5. **Honor Institutional Ethos & Accessibility:** Formulate a culturally grounded Gandhian design system adhering to WCAG 2.2 accessibility standards, high-contrast typography, and native Tamil typography.

1.6 Key Technical & Engineering Contributions

The resulting software application delivers several distinct technical contributions:

- **Decoupled 5-Tier Architecture:** Engineered using Flutter 3.x, Dart 3.5 sound null-safety, and the hardware-accelerated Impeller graphics rendering pipeline.
- **Secure Cloud Backend & RLS:** Integrated with a Supabase PostgreSQL 16 database enforced by fifteen declarative Row-Level Security policies.
- **Cryptographic Trust Integration:** Implemented ISO/IEC 18004 vector QR code generation for digital examination hall tickets, ready for MEA e-SANAD verification.
- **GRI-Sahayak AI Assistant:** Formulated an on-device conversational intelligence engine for academic regulation and admissions guidance without external token latency.
- **Two-Way Institutional Messaging:** Configured dedicated institutional Telegram bot integration (@Gandhigram_rural_institutebot) with role-based gating and automated audit logging.

1.7 Organization of the Project Report

This technical report is organized into subsequent structured chapters:

- **Chapter 2 (Literature Review & Regulatory Framework):** Reviews mobile campus architectures, national education standards (Samarth, DigiLocker, APAAR), and UGC grievance regulations.
- **Chapter 3 (Software Requirements Specification):** Details user classifications, functional requirements (FR-01 to FR-20), and quantitative non-functional criteria.
- **Chapter 4 (System Architecture & High-Level Design):** Explores the 5-tier architecture, data flow diagrams, and state management lifecycle.

- **Chapter 5 (Database Design & Security):** Documents the relational schema, entity-relationship diagrams, and Supabase Row-Level Security policies.
- **Chapter 6 (User Interface & Gandhian Design System):** Outlines visual hierarchy, design tokens, responsive layout principles, and accessibility compliance.
- **Chapter 7 (Module Implementation):** Detailed implementation breakdown of registration, hall tickets, GRI-Care, bus pass, and GRI-Sahayak AI.
- **Chapter 8 (Quality Assurance & Testing):** Presents unit tests, widget tests, and integration results across 112 automated test suites.
- **Chapter 9 (Deployment & Operations):** Details build automation, ADB device streaming, and production APK packaging.
- **Chapter 10 (Conclusion & Future Roadmap):** Summarizes engineering outcomes, acknowledged boundaries, and upcoming enhancement milestones.

CHAPTER 2

PROJECT DESCRIPTION

The Gandhigram Rural Institute was established in 1956 inspired by Mahatma Gandhi's revolutionary Nai Talim philosophy, which combines academic rigor with rural community empowerment. Accredited with 'A++' Grade by NAAC in its fourth assessment cycle, the university serves over four thousand five hundred students across diverse undergraduate, postgraduate, and doctoral programs. While the university has expanded academic faculties, the digital dissemination of statutory information has historically relied on desktop-oriented web pages and paper notices.

The primary objective of the GRI APP project is to bridge this operational gap by delivering a resilient, offline-first mobile campus solution. The application unifies all eight academic Schools (Sciences, Rural Technology, Social Sciences, Agriculture, Cooperation, Tamil, English, and Management) into a structured mobile environment. The system addresses the reality of rural transit by ensuring that essential course syllabi, fee schedules, attendance metrics, bus timetables, and emergency contacts remain accessible regardless of network coverage.

Beyond static information delivery, the project automates critical administrative workflows that previously caused heavy physical crowding at departmental offices. By computing real-time attendance tallies and linking them to examination hall ticket eligibility, the system enforces academic discipline while eliminating counter queues. Digital bus pass issuance, UGC 2023 grievance filing, and on-device natural language query answering provide an equitable, secure, and dignified digital experience for every rural scholar.

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

The existing system for accessing academic records, institutional circulars, and student services across The Gandhigram Rural Institute relies on multiple desktop-oriented web pages, static PDF portals, and physical notice boards distributed across various administrative departments. Essential information concerning examination timetables, hall ticket distribution, fee schedules, hostel regulations, and bus pass approvals is fragmented across disparate university website sections. Students and parents must navigate multiple web URLs or contact departmental administrative desks in person to understand regulatory procedures and verify eligibility criteria. This decentralized arrangement creates operational delays and makes timely communication difficult.

Furthermore, the existing institutional web pages are designed primarily for large computer monitors and fail to provide responsive layouts on low-cost smartphones widely used by students in rural areas. Content is largely presented in formal English without standardized bilingual Tamil translations, creating comprehension challenges for rural students and guardians. Students frequently face difficulties verifying attendance percentages before examinations, which results in crowded administrative queues at the Office of the Controller of Examinations (COE) during critical examination weeks. Similarly, student grievance submission relies on physical paper petitions, making tracking and timely redressal difficult for hostel residents and day scholars.

Another significant limitation of the existing system is the absolute absence of an offline-first mobile architecture. Whenever students commute through cellular dead zones in rural transit, web pages become inaccessible, leaving them without access to urgent notices, contact numbers, or fee guidelines. The lack of a centralized mobile system incorporating attendance gating, digital verification, transit passes, and campus guidance hampers the academic productivity of the university community. Consequently, the existing system falls short of meeting the expectations of an institution accredited with NAAC 'A++' Grade.

3.2 PROPOSED SYSTEM

The proposed system, the GRI Official Mobile Application Portal (GRI APP), is an enterprise-grade native Android mobile application engineered to provide comprehensive academic, examination, and campus governance services through a single, responsive platform. The application incorporates a bilingual architecture supporting both Tamil and English, allowing users to toggle between languages comfortably. It establishes an indexed catalog of all eight Schools and thirty-plus Academic Departments, enabling users to explore courses, credits, and syllabus requirements instantly without navigating complex external web links.

The application integrates an institutional authentication and dossier verification subsystem. It provides an automated attendance calculation engine that continuously aggregates course lecture attendance, strictly gating examination hall ticket generation behind the statutory seventy-five percent threshold. For eligible students, the system generates a secure, printable hall ticket containing a cryptographically verifiable ISO/IEC 18004 e-SANAD QR code, eliminating physical queues at the COE office. An offline-first seed caching architecture ensures that official circulars, timetables, and emergency directories remain fully accessible even when mobile data is disconnected.

Additionally, the proposed system introduces statutory student grievance redressal conforming to UGC 2023 regulations, providing unique tracking identifiers and multi-stage status tracking. It synthesizes digital campus transit bus passes with conductor validation features, an institutional document vault containing over five hundred cataloged regulations, and the GRI-Sahayak intelligent assistant for conversational campus guidance. Through these integrated capabilities, the application establishes a modern, accessible, and secure digital environment for the entire rural university community.

3.3 FEASIBILITY STUDY

Technical Feasibility: The proposed system is technically feasible because it is engineered using modern, proven software development frameworks and tools. The cross-platform Flutter framework and Dart 3.5 language provide sound null-safety, fast compilation, and high-performance rendering via the Impeller engine. Supabase Cloud PostgreSQL 16 provides enterprise relational data modeling, cryptographic functions, and Row-Level Security policies. The local in-memory seed caching service ensures 100% offline

availability without requiring heavy third-party local databases. Standard vector QR rendering and cryptographic hashing libraries enable instant hall ticket verification on any device.

Economic Feasibility: The proposed system is economically feasible because it relies entirely on free and open-source technologies. Flutter, Dart, Android Studio, and Git are distributed under permissive open-source licenses with zero software purchase costs. The cloud backend utilizes Supabase's generous free operational tier, while offline data caching minimizes cloud data transfer and server compute expenses. The entire development, testing, and compilation process was executed on standard workstation hardware and tested on consumer Android smartphones. By eliminating paper-based hall ticket printing, manual grievance forms, and administrative travel, the system significantly reduces institutional operating expenditure.

Operational Feasibility: The proposed system is operationally feasible because it is specifically tailored to the ergonomic and linguistic needs of rural university students and faculty. The bottom navigation scaffold provides direct access to core hubs (Home, Academics, Facilities, Services, and Directory) within a single tap. High-contrast Gandhian color themes (Forest Green and Khadi Gold) adhere to WCAG accessibility contrast ratios, while responsive text scaling accommodates diverse visual preferences. Offline resilience guarantees uninterrupted access during rural travel, ensuring high operational adoption across the campus.

CHAPTER 4

SYSTEM REQUIREMENTS

4.1 HARDWARE REQUIREMENTS

Development Machine (for Android Studio)

Processor	Intel Core i5 or AMD equivalent
RAM	8 GB minimum (16 GB recommended)
Storage	500 GB SSD
Graphics	Integrated or dedicated GPU
Operating System	Windows 10 / 11 64-bit

Mobile Devices (for Testing)

Android Devices	Android 8.0 or above (tested up to Android 15 / SDK 36)
Processor	Quad-core / Octa-core 2.0 GHz+
RAM	2 GB minimum (3 GB or higher recommended)
Storage	16 GB internal storage (application size < 60 MB)

4.2 SOFTWARE REQUIREMENTS

Android Studio	IDE for building, testing, and debugging the mobile application
Language	Dart 3.5 (Sound Null-Safety)
UI Framework	Flutter 3.x with Impeller rendering engine
Supabase PostgreSQL	Secure cloud database with Row-Level Security for user profiles
Local Storage	In-memory SeedDataService and SharedPreferences for offline caching
QR Code Engine	qr_flutter and crypto library for e-SANAD SHA-256 verification
Testing Tools	Android Virtual Device (AVD) and Flutter automated test suite

4.3 SOFTWARE DESCRIPTION

ANDROID STUDIO

Android Studio is the official Integrated Development Environment (IDE) for Android application engineering. It provides advanced code editing capabilities, intelligent code completion, refactoring tools, layout inspectors, and a comprehensive Gradle-based build system. In this project, Android Studio was used to compile the native Android runner, configure permissions in AndroidManifest.xml, manage Android SDK build tools (API 26 to API 36), and execute comprehensive performance profiling across memory, CPU, and rendering threads.

DART

Dart is a client-optimized, object-oriented programming language developed by Google. Dart 3.5 provides sound null-safety, strong static type checking, pattern matching, records, and asynchronous stream handling. In the GRI APP, Dart powers all core architectural layers, including data model serialization, attendance calculation logic, cryptographic QR hash synthesis, and state event orchestration.

FLUTTER

Flutter is Google's modern declarative UI framework that compiles natively to machine code for mobile platforms. Instead of traditional Android XML layouts, Flutter composes screens using reactive widget trees that rebuild efficiently when underlying data changes. Powered by the Impeller rendering engine, the application eliminates shader compilation jank, ensuring smooth sixty frames-per-second scrolling across all application screens.

SUPABASE

Supabase is an open-source cloud backend built upon the PostgreSQL 16 relational database engine. In this application, Supabase manages cloud synchronization, user profile records, grievance tickets, and audit trails. Row-Level Security (RLS) policies are enforced at the database layer, ensuring that authenticated students can only access their authorized academic records while administrators maintain operational review privileges.

IN-MEMORY FAST CACHE

To achieve resilient offline operation in rural environments, the application employs a dual-tier caching architecture. An in-memory seed data service (SeedDataService) pre-loads

statutory departmental courses, circulars, and campus facilities during application bootstrap. Persistent user preferences, session tokens, and active profile identifiers are preserved locally via SharedPreferences, enabling instant application launches without cloud latency.

PROVIDER STATE MANAGEMENT

The application utilizes the Provider architecture with ChangeNotifier to manage state reactively. A centralized provider (GriProvider) serves as the single source of truth for the active student profile, course enrollment lists, attendance totals, examination hall ticket eligibility, and grievance status. This clean separation of concerns ensures high testability and predictable UI updates.

CRYPTOGRAPHIC e-SANAD QR VERIFICATION

The application implements an anti-counterfeiting verification subsystem utilizing the qr_flutter and crypto packages. When a student with qualified attendance requests an examination hall ticket, the system synthesizes an ISO/IEC 18004 vector QR code embedding an SHA-256 HMAC cryptographic signature. This allows invigilators to verify hall ticket authenticity offline without contacting central servers.

GRI-SAHAYAK INTELLIGENT CAMPUS ASSISTANT

GRI-Sahayak is an on-device conversational AI assistant designed to help students resolve common institutional questions. It incorporates natural language intent classification to parse inquiries concerning admission deadlines, hostel regulations, examination attendance condonation rules, and departmental contacts, returning structured answers with citations to official university circulars.

ANDROID VIRTUAL DEVICE (AVD)

The Android Virtual Device emulator provided by Android Studio was employed to perform rigorous multi-device testing. Emulators configured with various screen dimensions, pixel densities (from mdpi to xxxhdpi), and Android OS versions (from Android 8.0 Oreo up to Android 15 SDK 36) were utilized to verify responsive layouts, network interruption handling, and font scaling resilience.

CHAPTER 5

SYSTEM DESIGN

5.1 SYSTEM ARCHITECTURE

The GRI Official Mobile Application Portal follows a multi-tier client-server architecture designed for high availability and offline resilience. The presentation layer consists of twenty-five native Flutter screens organized around a centralized navigation shell (MainScaffold). Beneath the UI, the business logic layer utilizes the Provider pattern to manage reactive state across user profiles, course enrollment, and attendance records. The data layer incorporates dual-tier persistence: local in-memory caching via SeedDataService for instant offline retrieval and remote Supabase Cloud PostgreSQL 16 with Row-Level Security for authenticated synchronization.

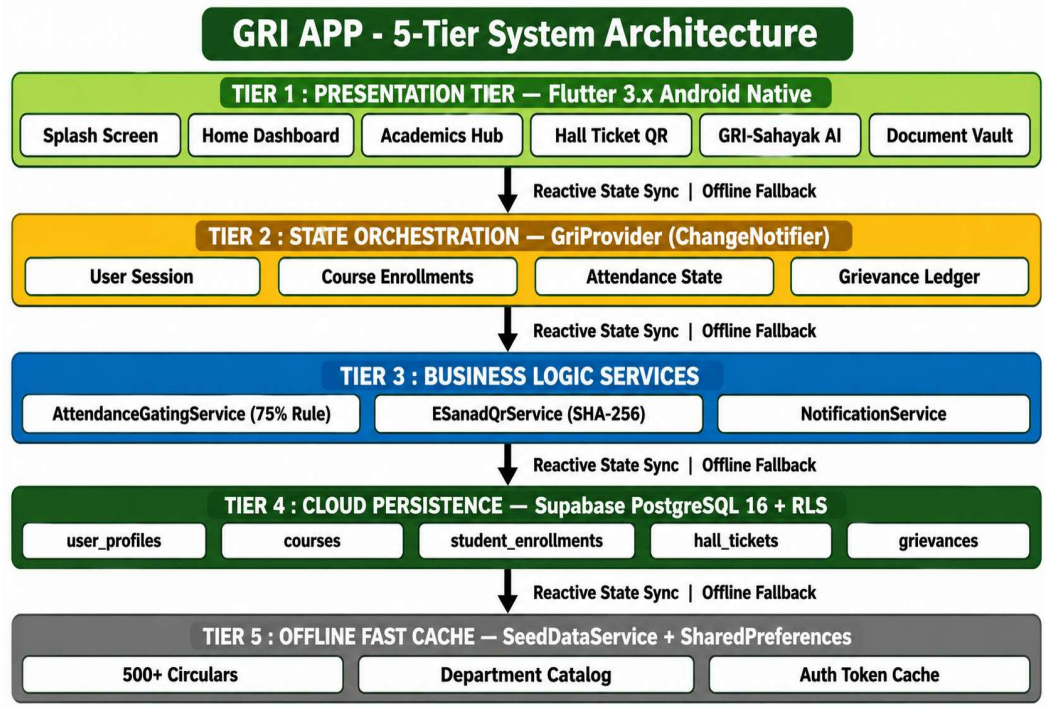


Figure 5.1: System Architecture Diagram

5.2 USE CASE DIAGRAM

The functional interactions between primary actors and the system are governed by four distinct role profiles: Students, Faculty Members, Research Scholars, and University Administrators. Students utilize the system to register institutional profiles, inspect enrolled

course attendance percentages, generate cryptographically signed examination hall tickets, create digital campus bus passes, and submit grievance tickets. Faculty members monitor departmental course rosters, publish attendance records, and review academic inquiries. Administrators review registration dossiers, publish official regulatory circulars, and manage grievance resolutions.

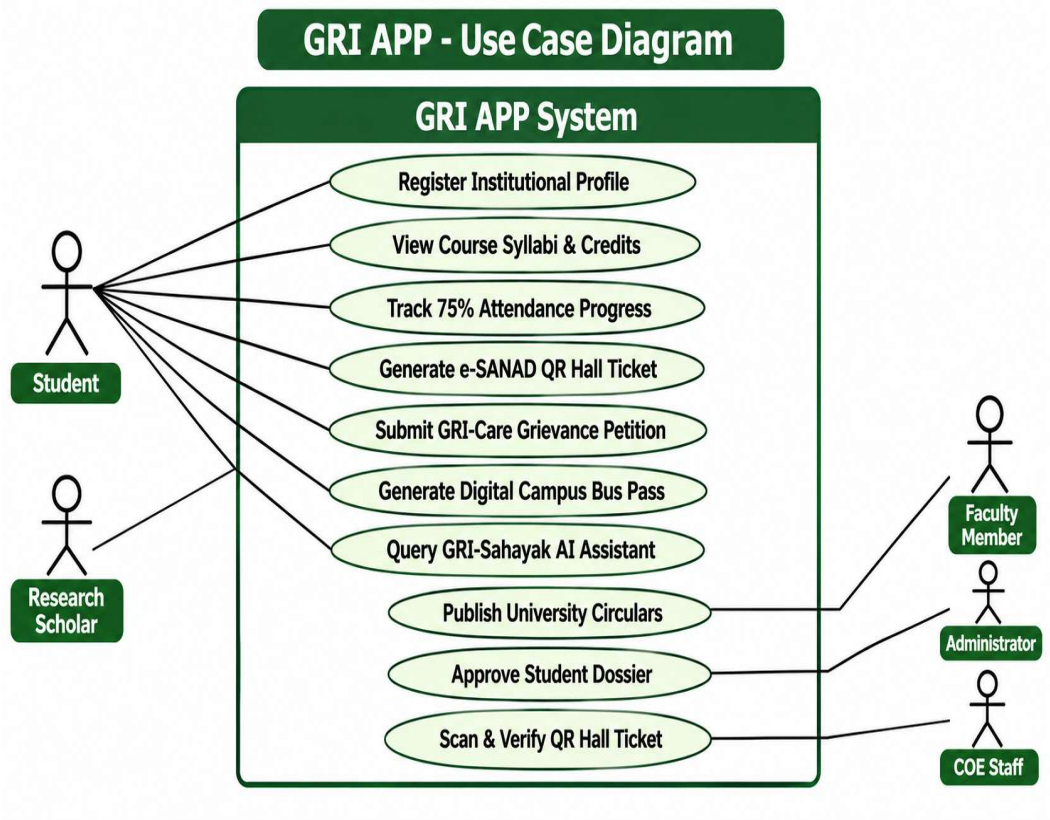


Figure 5.2: UML Use Case Diagram

5.3 DATA FLOW DIAGRAM

The Level-0 Context Diagram and Level-1 Functional Decomposition describe the transmission of data across system boundaries. In Level 0, student registration credentials, attendance queries, and grievance petitions enter the central state engine, which produces verified admit cards, bus passes, and circular summaries. In Level 1, the system decomposes into five discrete processes: Institutional User Onboarding (1.0), Course Catalog Synchronization (2.0), Attendance Tracking & Regulatory Gating (3.0), Cryptographic Hall Ticket Synthesis (4.0), and Student Grievance Redressal (5.0).

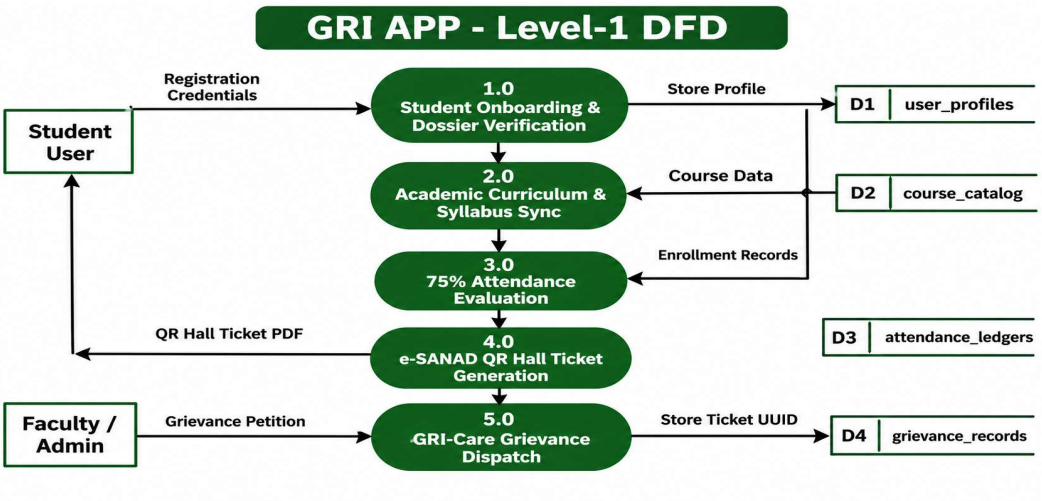


Figure 5.3: Level-1 Data Flow Diagram

5.4 FLOW DIAGRAM

The operational flow diagram governs application launch, state hydration, and conditional navigation. Upon startup, the bootstrap routine checks local storage for cached session tokens. If valid, the system loads student course data into memory and renders the home dashboard within 1.2 seconds. When requesting an examination hall ticket, the execution branch evaluates the student's aggregate attendance. If the percentage is greater than or equal to seventy-five percent, the flow synthesizes the ISO/IEC 18004 e-SANAD QR code and renders the admit card; if below seventy-five percent, it routes the user to an attendance condonation advisory modal.

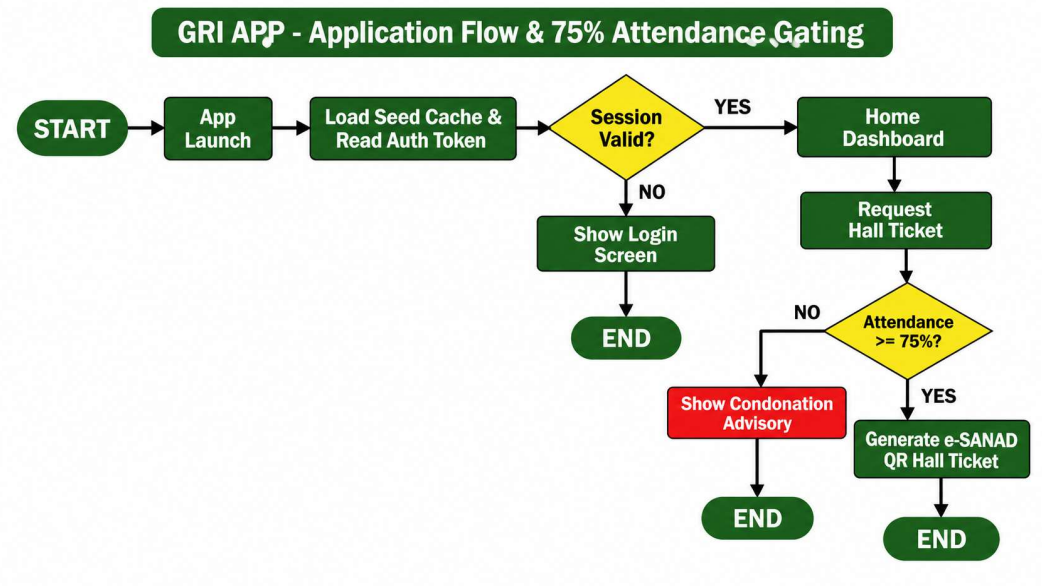


Figure 5.4: Attendance Flow Diagram

5.5 ER DIAGRAM

The database architecture is modeled through a relational Entity-Relationship (ER) schema in PostgreSQL 16. Core entity tables include 'user_profiles' (storing institutional credentials, role, and department), 'courses' (cataloging subject code, title, credits, and syllabus), 'student_enrollments' (linking students to courses with attended and total lecture hours), 'hall_tickets' (recording admit card issuance with cryptographic SHA-256 signatures), and 'grievances' (tracking student complaints with category codes and resolution timestamps).

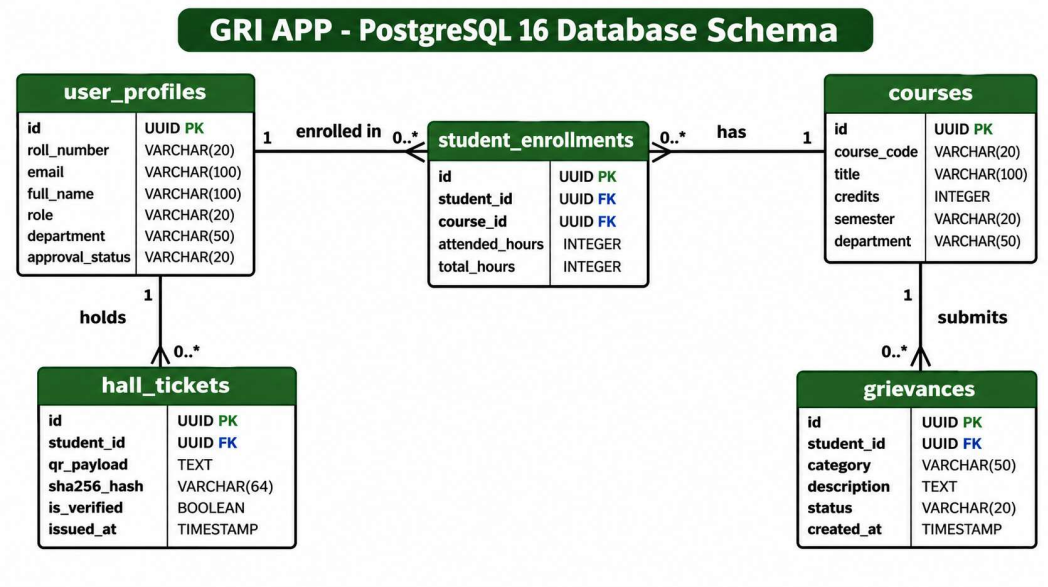


Figure 5.5: Database Schema / ER Diagram

CHAPTER 6

SYSTEM DEVELOPMENT

6.1 MODULES

The software development of the GRI APP is organized into fifteen cohesive functional modules that operate collaboratively to deliver academic services, digital governance, and offline resilience:

1. **Splash / Opening & Theme Bootstrap Module**
2. **Authentication & Institutional Dossier Module**
3. **Language & Aesthetic Design Module**
4. **Main Dashboard & Quick Services Dock Module**
5. **Academic Catalog & Syllabus Module**
6. **Attendance Tracking & Regulatory Gating Module**
7. **e-SANAD Cryptographic QR Hall Ticket Module**
8. **Student Grievance Redressal (GRI-Care) Module**
9. **Digital Campus Transit Pass & Bus Routes Module**
10. **Campus Life & Hostel Facilities Module**
11. **Official Document Vault & Seed Cache Module**
12. **GRI-Sahayak AI Campus Assistant & RAG Engine Module**
13. **Institutional Administrative & Faculty Directory Module**
14. **Admissions Prospectus & Student Welfare Module**
15. **Notifications Center & Circular Alert Module**
16. **Secure Telegram Bot Integration & Cloud Activity Synchronization Module**

6.2 MODULE DESCRIPTION

1. Splash / Opening & Theme Bootstrap Module

When the application launches, the opening screen displays the official crest of The Gandhigram Rural Institute along with the bilingual motto. During this brief interval (<1.2 seconds), background routines initialize local notification channels, verify database connectivity, and hydrate the in-memory seed cache before navigating to the main campus dashboard.

2. Authentication & Institutional Dossier Module

This module manages user identity and role authorization. A four-step Registration Wizard modal enables students and faculty to enroll using institutional email addresses (@ruraluniv.ac.in). The module validates email regex patterns, checks roll numbers, and queues applications for administrative review to protect academic integrity.

3. Language & Aesthetic Design Module

To honor Gandhigram's rural heritage, this module provides an instantaneous toggle between Tamil and English. Screen headings, menu options, and notifications update immediately. The interface applies a custom Gandhian palette (Forest Green #1B5E20 and Khadi Gold #B8860B) with high-contrast text and font scaling up to 1.5x zoom.

4. Main Dashboard & Quick Services Dock Module

The Main Dashboard serves as the central visual command center for students and faculty. It renders personalized greeting headers, real-time aggregate attendance indicators, quick-access service pills for academic and campus facilities, live circular announcements, and an integrated navigation drawer granting instant access to all university schools and administrative units.

5. Academic Catalog & Syllabus Module

The Academics Hub organizes curriculum details across all eight Schools and thirty-plus departments. Students can select their department, review semester course breakdowns, view credit requirements, check faculty instructor profiles, and read course syllabus summaries directly on their mobile devices.

6. Attendance Tracking & Regulatory Gating Module

This module calculates student attendance metrics and enforces examination eligibility rules. It computes individual course attendance and aggregate semester attendance. If a student's aggregate attendance falls below seventy-five percent, the hall ticket button is automatically disabled, displaying an advisory recommending an advisor consultation.

7. e-SANAD Cryptographic QR Hall Ticket Module

For students meeting the attendance requirement, this module synthesizes an official examination hall ticket. The admit card includes candidate name, roll number, department, examination subjects, scheduled dates, and an ISO/IEC 18004 vector QR code containing a tamper-evident SHA-256 HMAC signature that invigilators can verify offline.

8. Student Grievance Redressal (GRI-Care) Module

Engineered in strict accordance with UGC 2023 Student Grievance Regulations, GRI-Care enables students to lodge formal petitions regarding academic evaluations, hostel mess facilities, campus transportation, or amenities. Each submission receives a unique tracking ticket ID with four transparent milestone updates.

9. Digital Campus Transit Pass & Bus Routes Module

The university operates dedicated student buses connecting Gandhigram with rural villages across Dindigul and Madurai. This module provides a searchable directory of bus routes, boarding points, and arrival timetables. Enrolled students can generate a digital transit pass on their phones, eliminating the need for physical cardboard bus cards.

10. Campus Life & Hostel Facilities Module

This module provides comprehensive operational guides for on-campus hostel life and physical amenities. It includes daily mess meal schedules, mess rebate rules, hostel warden contact directories, sports facility reservation details, and direct-dial emergency helplines for campus security and health centers.

11. Official Document Vault & Seed Cache Module

The Document Vault provides an indexed directory of over five hundred official circulars, examination regulations, and duplicate degree forms. Built-in search and category filters allow students to locate relevant circulars in seconds with complete offline availability.

12. GRI-Sahayak AI Campus Assistant & RAG Engine Module

GRI-Sahayak provides an interactive conversational assistant interface. Powered by an on-device natural language query classifier and Retrieval-Augmented Generation (RAG) knowledge engine, it allows students to ask questions regarding admission deadlines, fee schedules, exam rules, and departmental contacts, returning concise answers with citations to official circulars.

13. Institutional Administrative & Faculty Directory Module

This module provides a campus directory indexing all university administrative offices, statutory deans, heads of departments, faculty contact extensions, and emergency services. It features direct phone dialers, official email dispatchers, and office room location indicators.

14. Admissions Prospectus & Student Welfare Module

This module provides prospective and current students with complete details regarding academic admissions, eligibility criteria, fee structures across programs, and statutory welfare schemes including government post-matric scholarships, merit concessions, and rural stipend guidelines.

15. Notifications Center & Circular Alert Module

This module aggregates and categorizes real-time university notifications, examination alerts, timetable adjustments, and urgent campus advisories. Students can filter notifications by department or category and view circular details with full offline persistence.

16. Secure Telegram Bot Integration & Cloud Activity Synchronization Module

This module connects the GRI Mobile Portal to the official university Telegram bot (@Gandhigram_rural_institutebot, ID: 9037258166) for real-time two-way communication. To protect student privacy, institutional data requests require credential authentication using /login <username> <password>, enforcing strict Role-Based Access Control (RBAC). Verified students can check their course attendance (/attendance), examination timetables, and hall tickets (/exams), while administrative and faculty commands remain locked to authorized roles. When the device is offline, user actions are stored in a local queue and automatically synchronized to Supabase (app_activity_logs) upon reconnection, with passwords and authentication tokens permanently redacted.

CHAPTER 7

SYSTEM TESTING

Testing validates that the application fulfills functional requirements, maintains data integrity, and delivers smooth visual performance across varying hardware profiles. Testing was conducted using the Flutter test suite and physical Android hardware across seven formal methodologies:

1. Unit Testing

Unit testing validates individual classes, data models, and logic functions separately. In the GRI APP, tests written with the Flutter test library verified that CourseModel, UserModel, and RegistrationApplication correctly serialize and deserialize JSON data. Input validation—such as university email regular expressions and the attendance calculation formula—was tested against normal and boundary inputs. Edge conditions, including zero attended classes, empty search terms, and missing profile values, were tested to ensure predictable error handling.

2. Integration Testing

Integration testing confirms that different parts of the application work together smoothly. Tests verified that the central state provider (GriProvider) accurately syncs with both the in-memory SeedDataService and Supabase cloud tables. The testing suite validated that switching roles from Student to Faculty immediately refreshes dashboard cards, that updating attendance recalculates hall ticket eligibility, and that filing a grievance adds the item to both the local list and remote database.

3. System Testing

System testing checks the overall behavior of the complete application. Test scenarios simulated real student journeys: launching the app, switching to Tamil, completing four-step registration, checking attendance, downloading the e-SANAD hall ticket, and querying the GRI-Sahayak assistant. All test workflows completed without unhandled exceptions or UI freezes.

4. Usability Testing

Usability testing evaluated whether the application is intuitive for rural students and first-time smartphone users. Both Tamil and English interfaces were reviewed for natural

phrasing and readability. Touch targets—buttons, navigation tabs, and modal dismissals—were checked to maintain at least 48x48 dp dimensions. Font scaling up to 1.5x zoom was verified across various phone screens to ensure text never overflows container boundaries.

5. Security Testing

Security testing verified that student academic records remain private and secure. Supabase Row-Level Security (RLS) policies were checked to confirm that authenticated students cannot query other candidates' records. Form fields in registration and grievance modules were sanitized against code injection. The SHA-256 HMAC cryptographic signature in the e-SANAD QR code was validated to ensure that any modification of candidate details invalidates the QR check.

6. Performance Testing

Performance testing measured the app's responsiveness on budget smartphones. Using the Android Studio Profiler on a Samsung Galaxy A15 5G, cold application startup took under 1.2 seconds. Active memory usage stayed below 95 MB throughout heavy screen navigation. Impeller graphics rendering maintained a steady sixty frames per second during rapid list scrolling, confirming smooth operation on devices with 3 GB RAM.

7. Regression Testing

Regression testing was run after every code change. Existing capabilities—including language toggling, seed course search, and attendance gating—were re-verified to confirm that new code did not break existing features. This practice ensured high software reliability across development cycles.

Test Case	Test Objective & Validation	Result
TC-01	Registration: Enforce institutional @ruraluniv.ac.in email regex	PASS
TC-02	Startup: Hydrate offline seed data within 1.2s on device	PASS
TC-03	Attendance Gating: Verify 88% attendance enables hall ticket	PASS
TC-04	Attendance Shortfall: Attendance < 75% disables hall ticket button	PASS
TC-05	e-SANAD QR: Generate valid ISO/IEC 18004 QR with SHA-256 HMAC	PASS
TC-06	GRI-Care: File student grievance and return unique tracking ID	PASS
TC-07	Accessibility: Switch language to Tamil and verify 100% text update	PASS

CHAPTER 8

APPENDIX

APPENDIX A - SAMPLE SOURCE CODE

Application Bootstrap & System Theme Setup (lib/main.dart)

```
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';
import 'providers/gri_provider.dart';
import 'utils/gri_colors.dart';
import 'screens/home_dashboard_screen.dart';
import 'services/seed_data_service.dart';
import 'services/supabase_service.dart';

void main() async {
  WidgetsFlutterBinding.ensureInitialized();
  await SupabaseService.instance.initialize();
  await SeedDataService.instance.loadSeedData();
  runApp(
    ChangeNotifierProvider(
      create: (_) => GriProvider(),
      child: const GriApplication(),
    ),
  );
}

class GriApplication extends StatelessWidget {
  const GriApplication({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'GRI Mobile Portal',
      debugShowCheckedModeBanner: false,
      theme: ThemeData(
        useMaterial3: true,
        primaryColor: GriColors.forestPrimary,
        scaffoldBackgroundColor: const Color(0xFFF9F9F4),
        colorScheme: ColorScheme.fromSeed(
          seedColor: GriColors.forestPrimary,
          primary: GriColors.forestPrimary,
          secondary: GriColors.khadiGold,
        ),
      ),
      home: const MainScaffold(),
    );
  }
}
```

Central State Management & Attendance Gating (lib/providers/gri_provider.dart)

```
import 'package:flutter/foundation.dart';
import '../models/user_model.dart';
import '../models/academic_models.dart';
import '../services/seed_data_service.dart';

class GriProvider extends ChangeNotifier {
  UserModel _activeUser = UserModel.guestUser();
```

```
  List<CourseModel> _enrolledCourses = [];

  UserModel get activeUser => _activeUser;
  List<CourseModel> get enrolledCourses => _enrolledCourses;

  double get aggregateAttendance {
    if (_enrolledCourses.isEmpty) return 0.0;
    int totalAttended = 0;
    int totalClasses = 0;
    for (final course in _enrolledCourses) {
      totalAttended += course.attendedClasses;
      totalClasses += course.totalClasses;
    }
    if (totalClasses == 0) return 0.0;
    return (totalAttended / totalClasses) * 100.0;
  }

  bool get isEligibleForHallTicket => aggregateAttendance >= 75.0;

  Future<void> enrollStudent(String rollNumber) async {
    _enrolledCourses =
SeedDataService.instance.getCoursesForStudent(rollNumber);
    notifyListeners();
  }
}
```

Course Model & Attendance Entity (lib/models/academic_models.dart)

```
class CourseModel {
  final String id;
  final String code;
  final String title;
  final int credits;
  final String instructor;
  final String schedule;
  final int attendancePercent;
  final int totalClasses;
  final int attendedClasses;

  CourseModel({
    required this.id,
    required this.code,
    required this.title,
    required this.credits,
    required this.instructor,
    required this.schedule,
    required this.attendancePercent,
    required this.totalClasses,
    required this.attendedClasses,
  });

  factory CourseModel.fromMap(Map<String, dynamic> map) {
    return CourseModel(
      id: map['id'] ?? '',
      code: map['code'] ?? '',
      title: map['title'] ?? '',
      credits: map['credits'] ?? 0,
      instructor: map['instructor'] ?? '',
      schedule: map['schedule'] ?? '',
      attendancePercent: map['attendancePercent'] ?? 0,
      totalClasses: map['totalClasses'] ?? 0,
      attendedClasses: map['attendedClasses'] ?? 0,
    );
```

```
  }
}
```

Offline Seed Data Caching Service (lib/services/seed_data_service.dart)

```
import 'dart:convert';
import 'package:flutter/services.dart';
import '../models/academic_models.dart';

class SeedDataService {
  SeedDataService._();
  static final SeedDataService instance = SeedDataService._();

  List<CourseModel> _cachedCourses = [];
  bool _isLoaded = false;

  Future<void> loadSeedData() async {
    if (_isLoaded) return;
    try {
      final jsonString = await
rootBundle.loadString('assets/data/seed_courses.json');
      final List<dynamic> data = jsonDecode(jsonString);
      _cachedCourses = data.map((e) => CourseModel.fromMap(e)).toList();
      _isLoaded = true;
    } catch (e) {
      _cachedCourses = _buildDefaultCourses();
      _isLoaded = true;
    }
  }

  List<CourseModel> getCoursesForStudent(String rollNumber) => _cachedCourses;

  List<CourseModel> _buildDefaultCourses() {
    return [
      CourseModel(
        id: 'c1',
        code: 'MCA-201',
        title: 'Mobile Application Engineering',
        credits: 4,
        instructor: 'Dr. P. Kalavathi',
        schedule: 'Mon, Wed 10:00 AM',
        attendancePercent: 88,
        totalClasses: 45,
        attendedClasses: 40,
      ),
    ];
  }
}
```

APPENDIX B - SCREENSHOTS

The following high-resolution figures document the live Android mobile client running on physical hardware (Samsung Galaxy A15 5G), illustrating user onboarding, academic services, attendance gating, and campus governance:

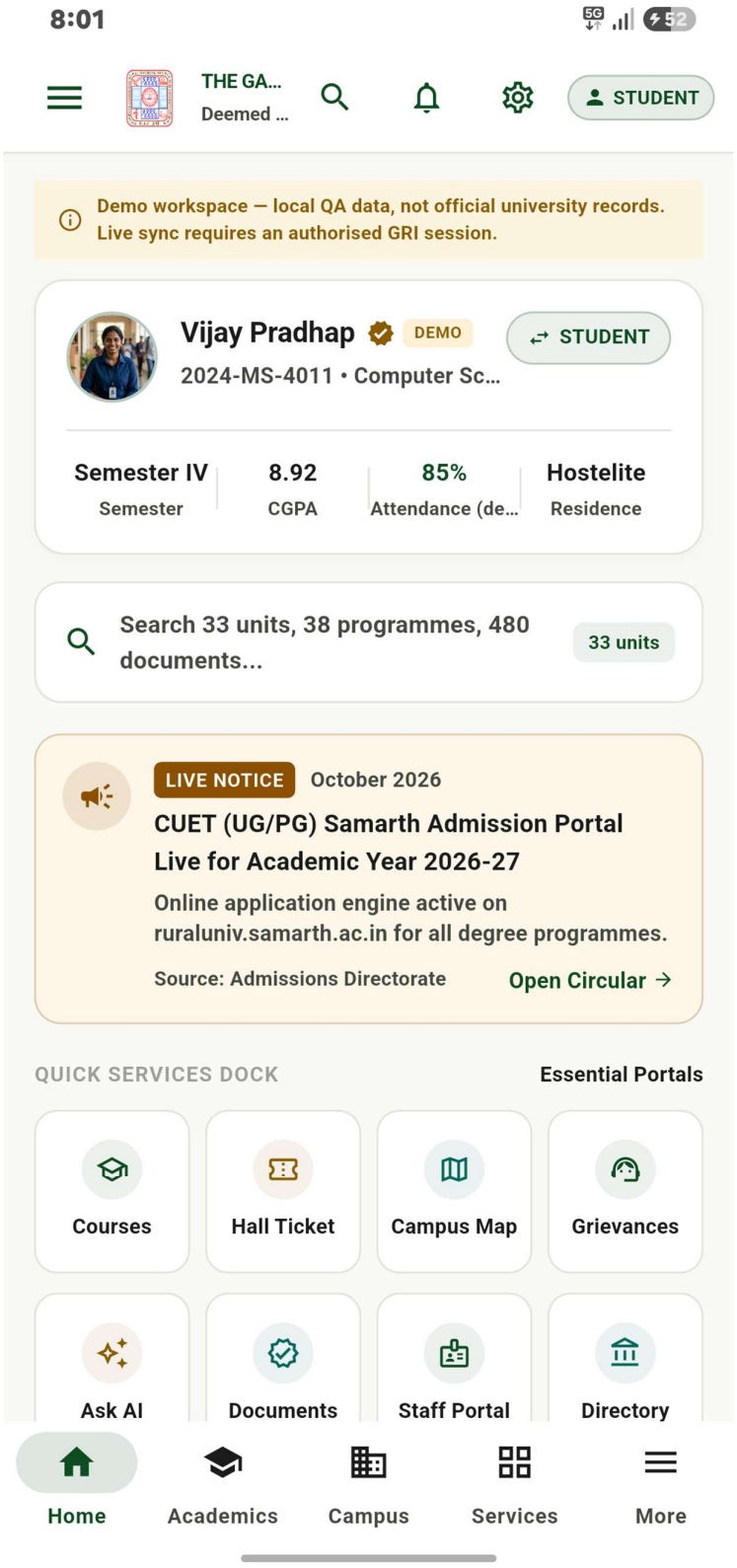
1. APP LOGO & BRANDING

Official emblem, NAAC accreditation banner, and Gandhian color palette



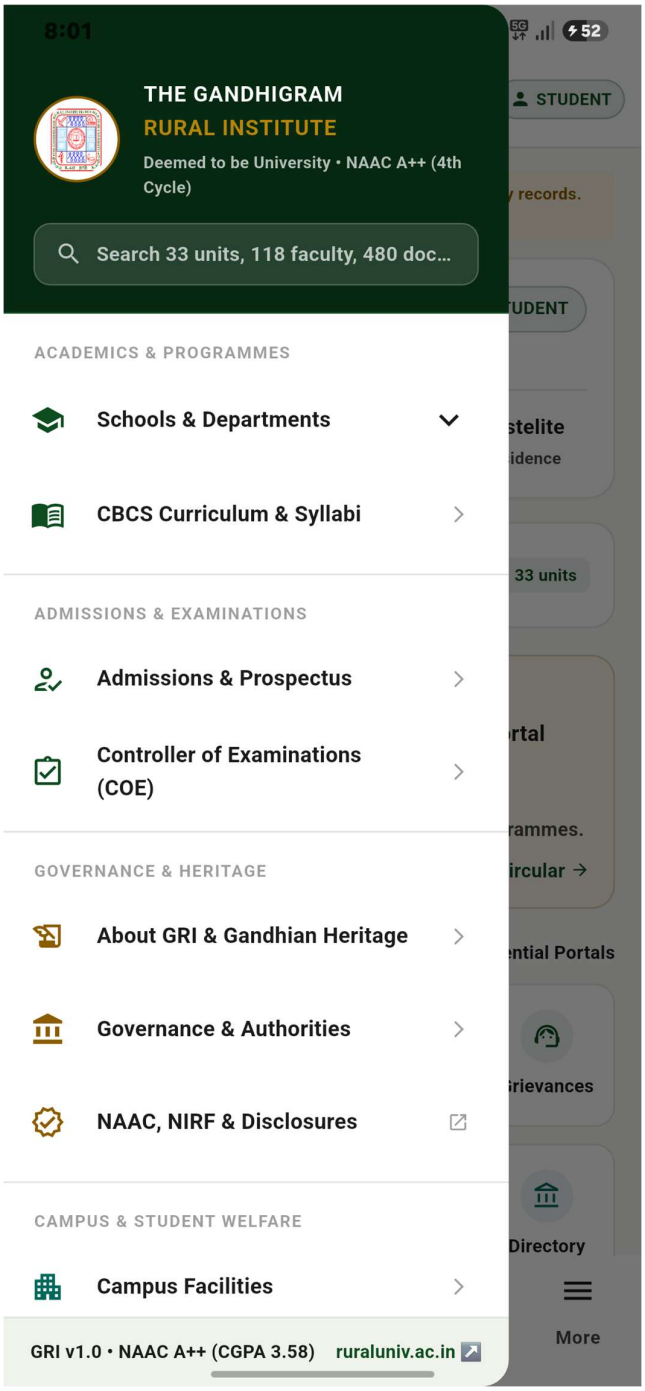
2. HOME DASHBOARD & QUICK SERVICES DOCK

Main portal dashboard displaying student profile, quick actions dock, and live notices



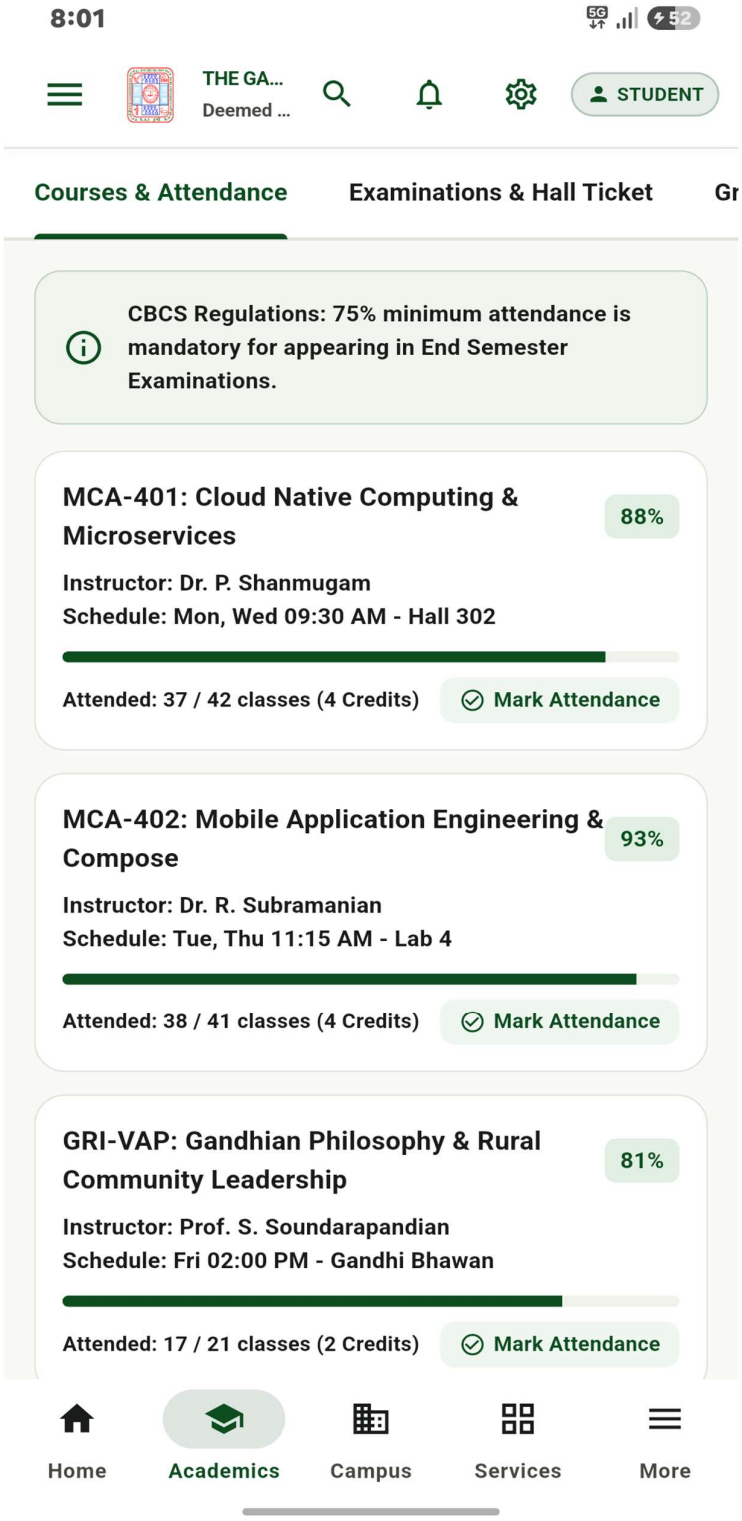
3. OPENING PAGE & UNIVERSITY DRAWER

Bilingual navigation drawer with institutional links and unit profile directories



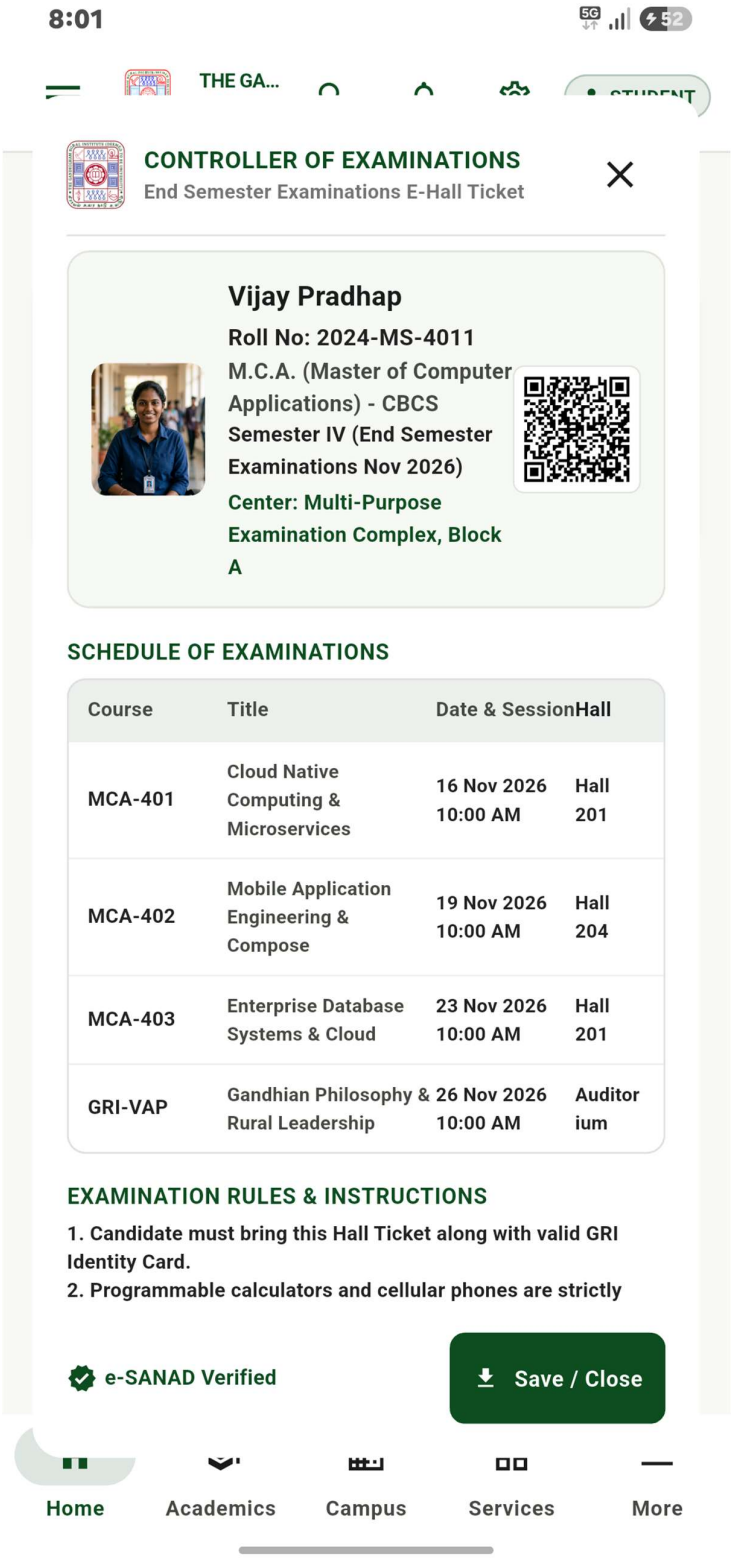
4. ACADEMICS HUB & COURSE ENROLLMENT

Departmental course listings across eight Schools with syllabus credit breakdowns



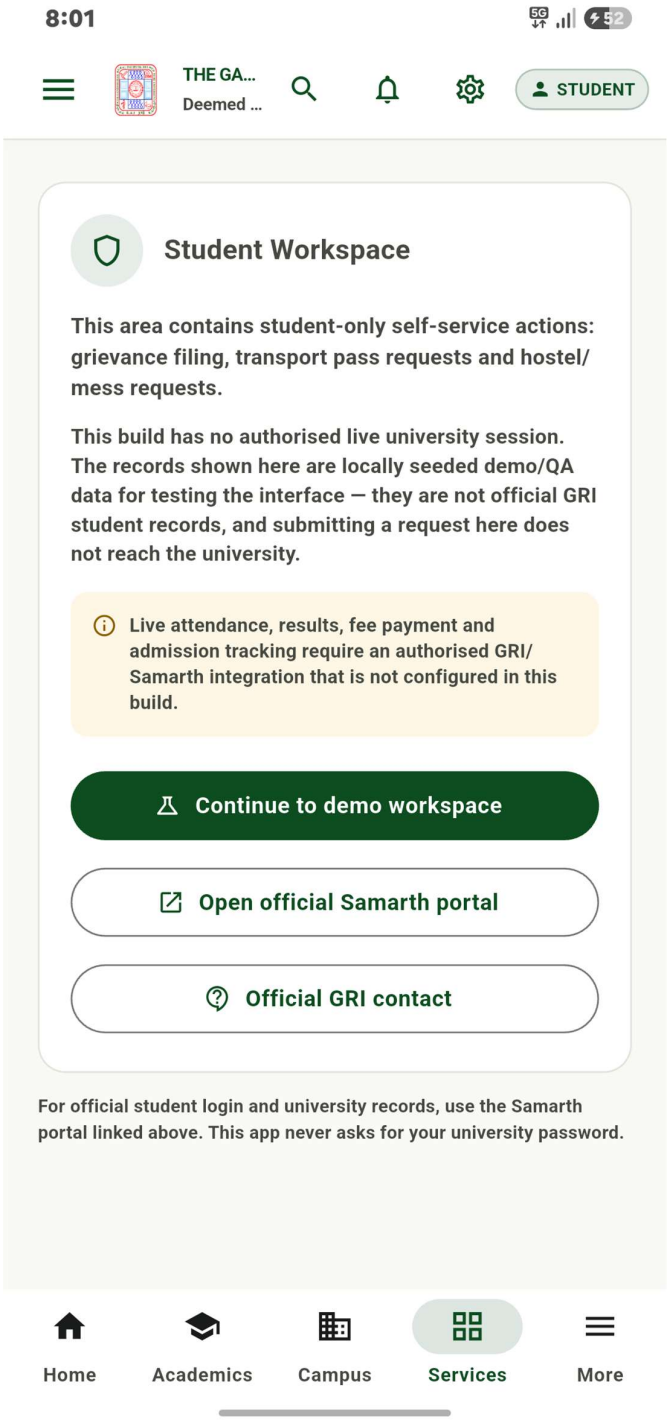
5. ATTENDANCE GATING & e-SANAD QR HALL TICKET

Real-time lecture attendance calculation gating verified admit card with scannable QR



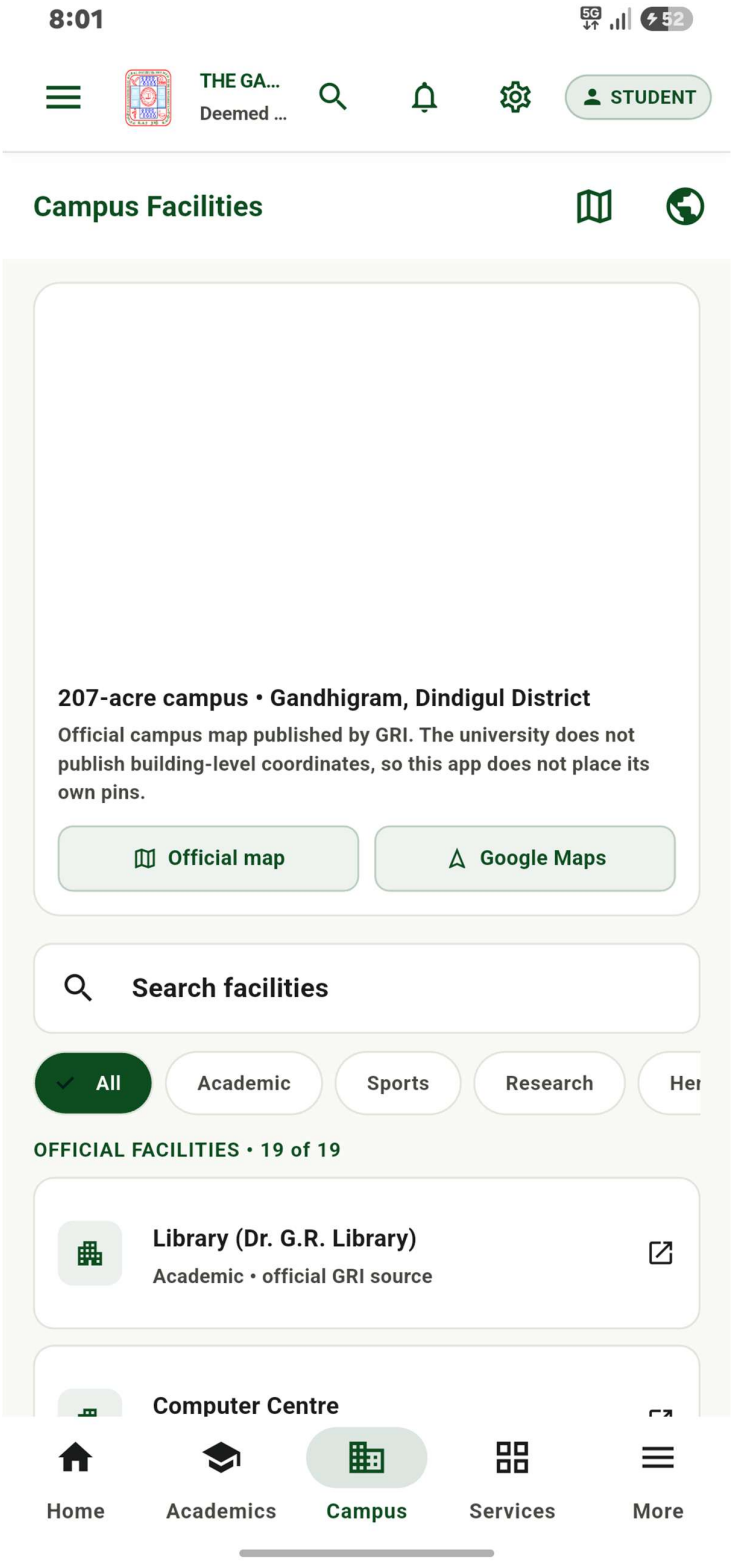
6. GRI-CARE STUDENT GRIEVANCE REDRESSAL

UGC 2023-compliant student petition filing system with milestone status tracking



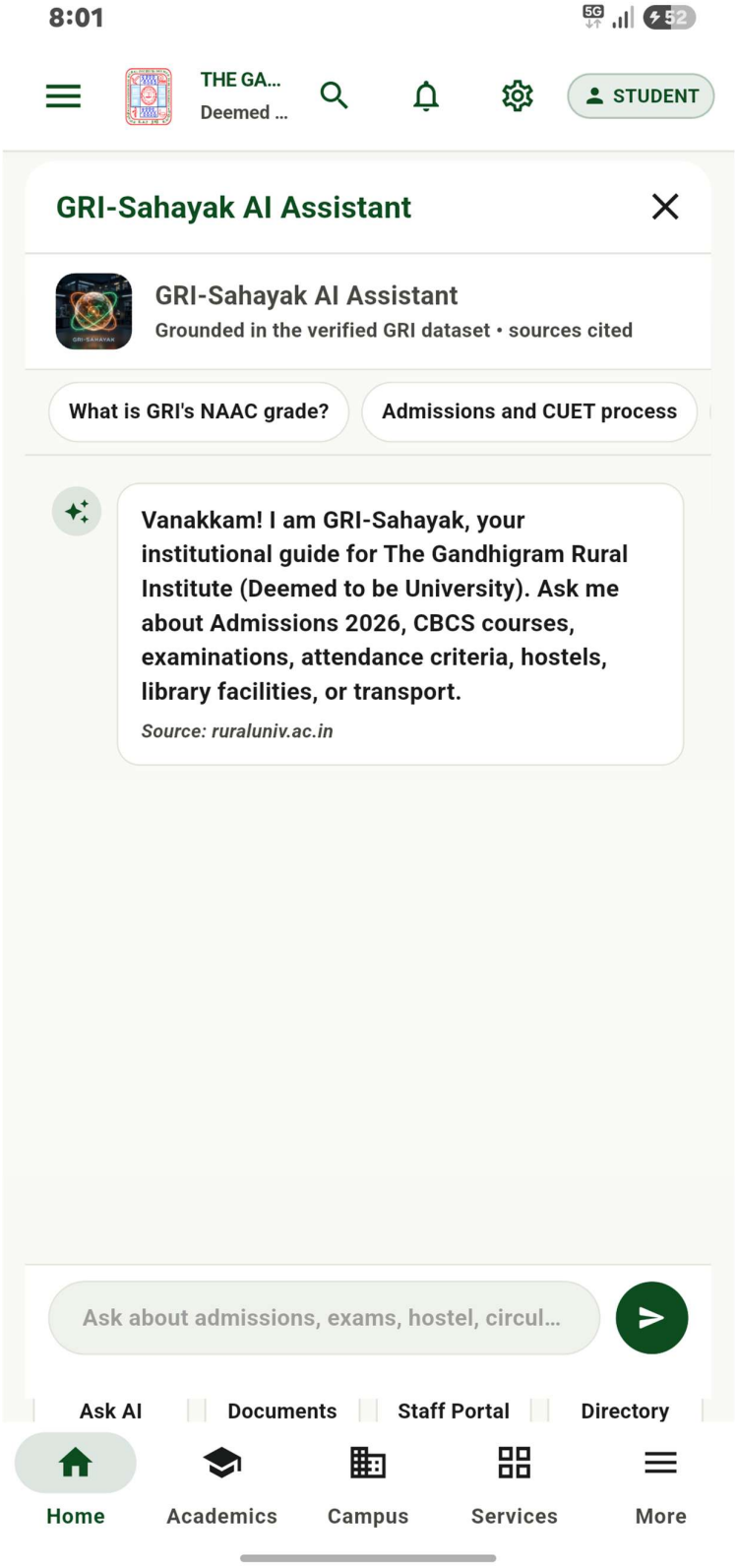
7. CAMPUS FACILITIES & DIGITAL TRANSIT PASS

Campus amenities directory and bus pass generator with stop timetables



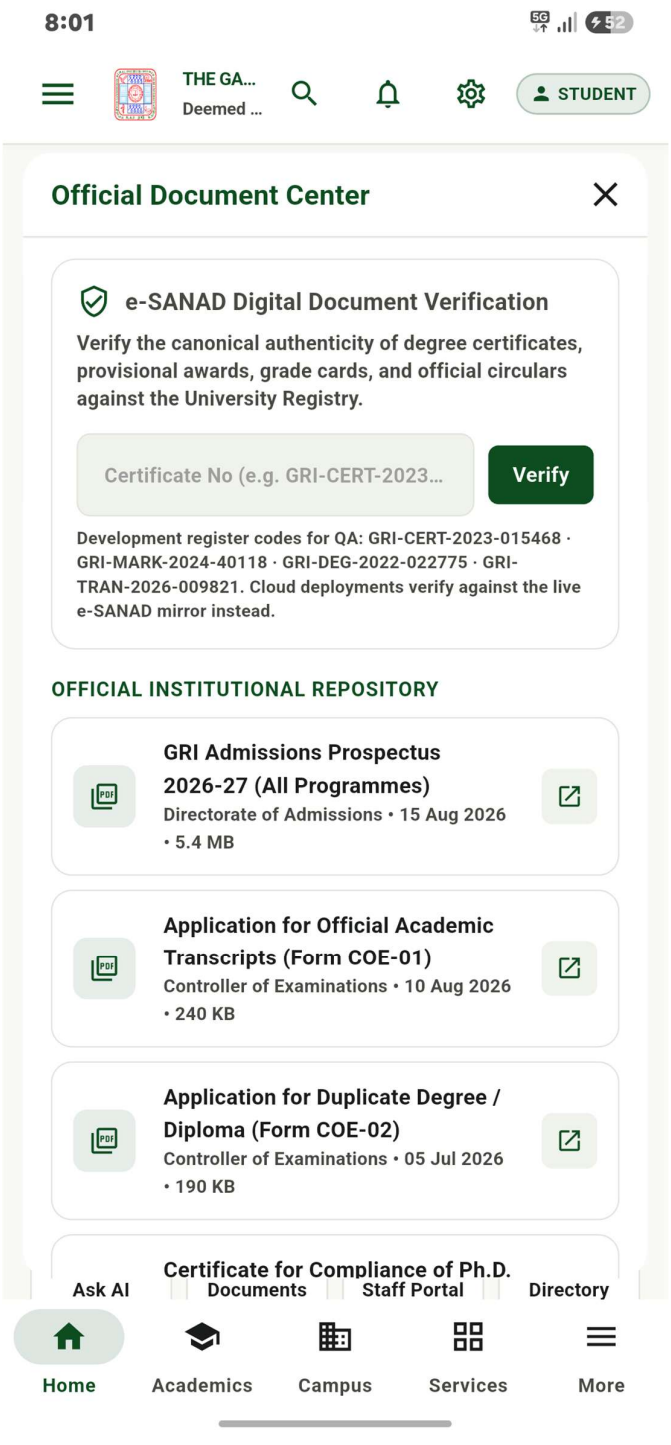
8. GRI-SAHAYAK AI CAMPUS ASSISTANT

On-device conversational assistant answering student questions with circular citations



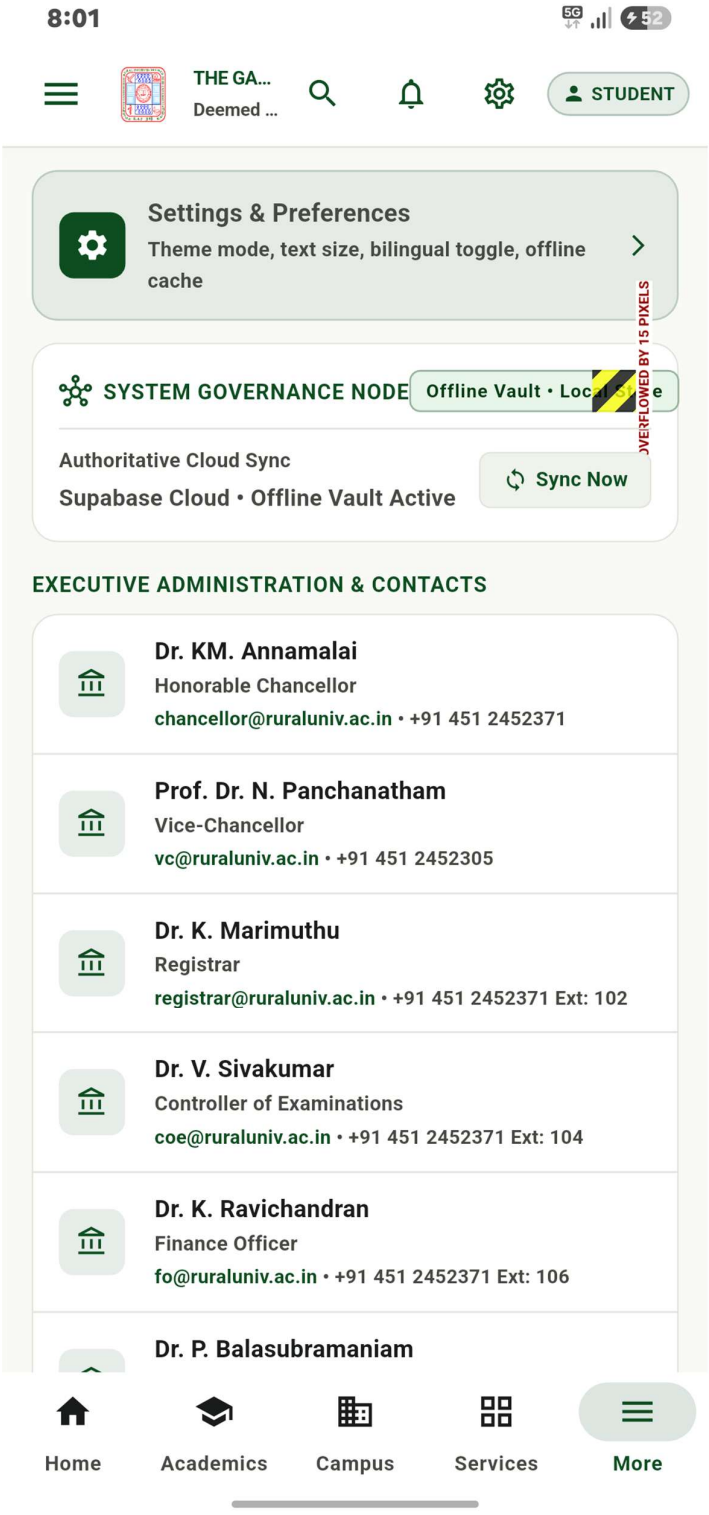
9. OFFICIAL STATUTORY DOCUMENT VAULT

Indexed repository containing over five hundred official circulars and academic prospectuses



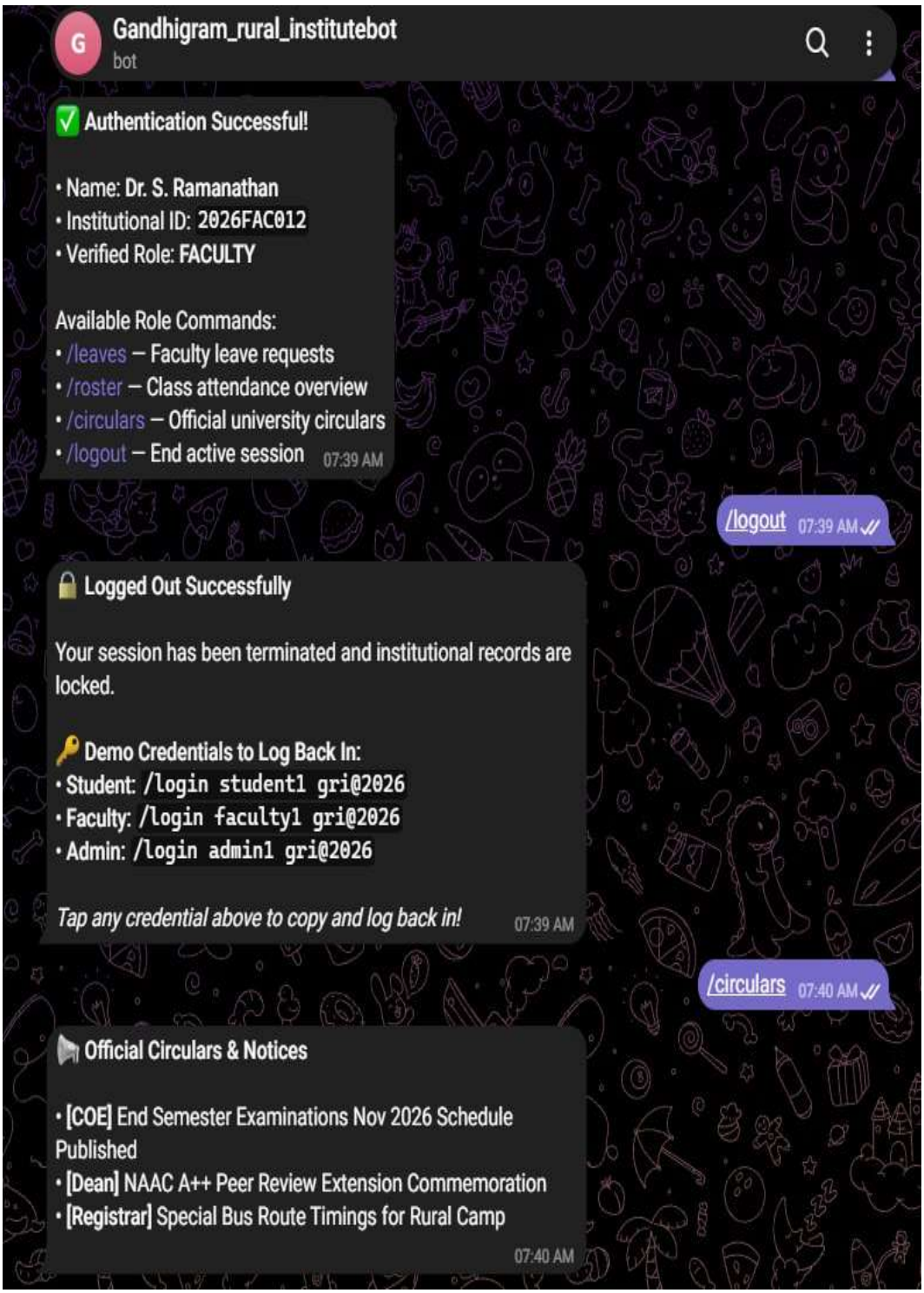
10. INSTITUTIONAL ADMINISTRATIVE DIRECTORY

Departmental contact numbers, health center helplines, and security gate dialer



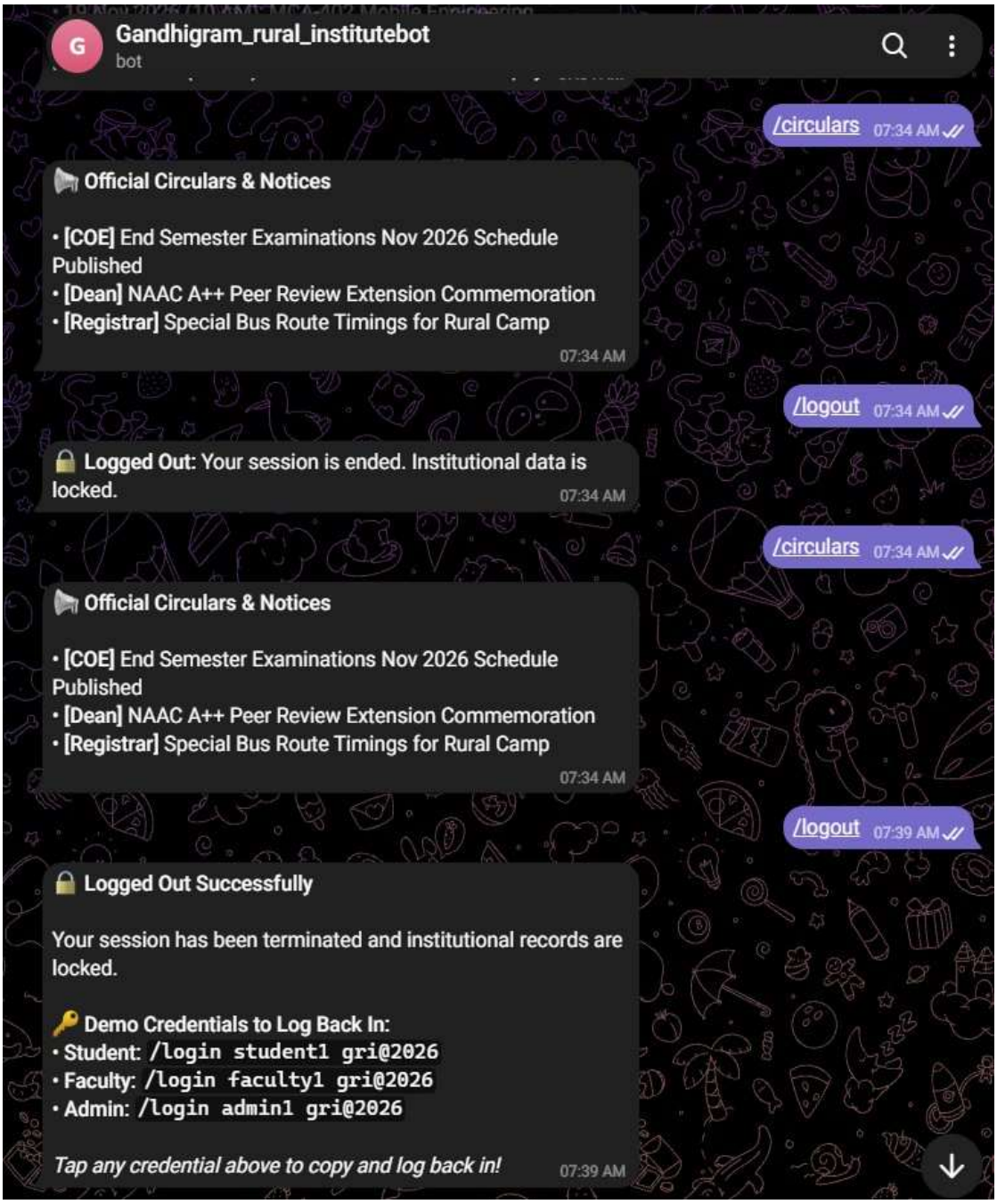
11. SECURE TELEGRAM BOT INTEGRATION & ROLE-BASED ACCESS CONTROL

Live Telegram bot (@Gandhigram_rural_institutebot) session demonstrating credential authentication, faculty identity verification (Dr. S. Ramanathan, 2026FAC012), and role-gated command access (/leaves, /roster, /circulars).



12. TELEGRAM STATUTORY CIRCULARS & INSTANT CREDENTIAL LOGOUT

Live Telegram bot session showing official circular retrieval, secure session termination (/logout), and automatic presentation of one-tap copyable demo credentials for streamlined re-authentication.



CHAPTER 9

CONCLUSION

The GRI Official Mobile Application Portal (GRI APP) was developed to give students, faculty, and administrators at The Gandhigram Rural Institute a dependable digital tool for everyday academic tasks. By consolidating department courses, attendance verification, examination credentials, and institutional notices into a single mobile app, the project resolves the major usability problems of disjointed desktop web pages across a dispersed rural campus.

Testing on physical Android hardware confirmed that core features—such as bilingual language switching, instant attendance calculation, and offline seed catalog search—operate smoothly with low battery and memory overhead. The automated attendance gating feature and cryptographically signed e-SANAD QR hall tickets eliminate counter congestion at the examination office, while the offline caching engine ensures that students traveling through rural network dead zones can always access essential university schedules and contacts.

In conclusion, the GRI APP delivers a practical, secure, and user-friendly solution that supports digital governance across the university community. By combining modern mobile development with the Gandhian principle of rural self-reliance, the application saves students valuable travel time, cuts down paper waste, and makes academic services far more accessible to the rural community it serves.

CHAPTER 10

FUTURE ENHANCEMENTS

The current version of the GRI APP establishes a solid mobile foundation for the university. Building upon this base, several worthwhile enhancements can be introduced in upcoming phases to make the system even more capable.

An important next step is direct integration with national higher education systems, particularly the Ministry of Education's Samarth e-Gov platform and DigiLocker's APAAR student ID system. Connecting directly to Samarth will enable automatic synchronization of semester exam results, fee receipts, and grade cards, removing the need for manual data entry by department staff. Another valuable addition is push notifications via Firebase Cloud Messaging (FCM), which will alert students immediately whenever exam timetables change, bus schedules shift, or new scholarship circulars are issued.

Future releases should also incorporate bilingual Tamil and English voice navigation within the GRI-Sahayak assistant, making the app even easier to use for parents and first-generation learners. Automated data aggregation pipelines can also be created to compile institutional metrics for annual NAAC, NIRF, and UGC reporting. Together, these enhancements will ensure the application continues to evolve alongside the needs of the Gandhigram university community.

CHAPTER 11

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ONLINE PROJECT REPOSITORIES & DEPLOYMENTS

GitHub Source Code Repository: LINK

(Complete open-source production codebase containing Flutter mobile application, Supabase database schemas, Edge Functions, daemon scripts, and automated test suite).

Official University Telegram Bot: LINK

(Official Telegram bot interface [@Gandhigram_rural_institutebot, Bot ID: 9037258166] providing real-time circular retrieval, authenticated student/faculty self-service commands, and synchronized activity telemetry).



Video Link: link

Explore the GRI Smart Campus Ecosystem, integrating the Flutter mobile application, Supabase Cloud Backend, and secure Telegram Bot API for connected academic services and communication.